

CIVICTRACKER — Hackathon Defense & Pitch Guide

Municipal Infrastructure Grievance & Accountability System • Judge Cheat-Sheet

1. THE 30-SECOND WINNING PITCH

"Current municipal complaint portals fail due to zero accountability, fake closures by contractors, and lack of citizen incentives. CivicTracker transforms civic grievance redressal with three pillars: (1) Dual-Stage Vision AI (Gemini) that validates defects on intake and audits Before/After repair photos with a strict binary YES/NO, (2) 15-Day Automated Statutory SLA with Point-in-Polygon Ward Routing, and (3) Civic Sense Gamification rewarding the Top 3 citizens every quarter with official Government Certificates."

2. SYSTEM ARCHITECTURE & TECH STACK

Frontend:	Next.js 15 App Router, React 19, Tailwind CSS	— Responsive UI, Portaled Modals, GIS Heatmap
AI Engine:	Google Gemini 2.0/1.5 Flash Vision + YOLOv8	— Defect validation, Severity 1-5, Before/After closure verification
Database & Auth:	Firebase Firestore & Google OAuth	— Realtime ticket sync, RBAC (Citizen vs. Dept Admin)
GIS & Spatial:	OSM Nominatim + Ray-Casting Polygon	— GPS reverse geocoding, instant ward polygon routing
SLA Engine:	Automated statutory timers & escalation	— 15-day countdown, 500-upvote emergency priority compression

3. CORE INNOVATION PILLARS

1. Dual-Stage Vision AI

Intake AI rejects non-defect photos/memes. Stage-2 Gemini audits Before vs After repair photos with binary YES/NO, preventing fake contractor closures.

2. 15-Day Statutory SLA

Strict statutory SLA countdown with automatic public overdue escalations to the Municipal Commissioner and emergency compression on 500 upvotes.

3. Point-in-Polygon Ward Routing

Mathematical ray-casting algorithm matches citizen coordinates directly against municipal ward polygons (PWD, Sanitation, Water Works) without human delays.

4. 3-Month Civic Honours Cycle

Permanent Civic Sense Rating (100-1000) preserved indefinitely, while competition points reset every quarter to award printable official Government Certificates of Merit.

4. TOP JUDGE QUESTIONS & BULLETPROOF ANSWERS

Q: How do you prevent spam or fake complaints?

A: Client HTML5 Canvas compresses images, and Gemini Vision evaluates the photo in ~800ms. If a non-defect image (like a meme or selfie) is uploaded, the API rejects it before any database entry is created.

Q: What stops contractors from uploading fake "fixed" photos?

A: Our Stage-2 AI Resolution Auditor sends both Before and After photos to Gemini Vision. If the pothole/garbage is still present or fix is incomplete, Gemini outputs a binary NO, blocking ticket closure.

Q: How does spatial routing work without manual forwarding?

A: We use a Ray-Casting point-in-polygon algorithm in zoneMatcher.ts that matches GPS coordinates against municipal polygon coordinates (ADMIN_ZONES) and dispatches directly to the assigned department.

Q: Why reset points every 3 months?

A: Resetting competition points quarterly allows new citizens an equal opportunity to compete and win government certificates, while permanent Civic Sense ratings (100-1000) are maintained for life.

5. 3-MINUTE LIVE DEMO SCRIPT FOR JUDGES

- [1]** Step 1 (Homepage /): Highlight live location detection (PHAGWARA), real-time ticket counters, and CT brand identity.
- [2]** Step 2 (Report /report): Upload a defect photo -> Show instant Gemini Vision defect classification and severity scoring.
- [3]** Step 3 (Resolver /department): Show proximity-sorted field tickets -> Upload resolution photo -> Trigger Gemini YES/NO audit.
- [4]** Step 4 (SLA Board /dashboard): Show the 10 official municipal wards ranked by speed and resolution rate (62% to 100%).
- [5]** Step 5 (Govt Honours /civic-score): Open Profile Modal -> Click Civic Score -> Open printable official Government Certificate!